EW-2 Project-2
Colour Sensor (Table No. 43)

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***Abstract*—The aim of this project is to design and build a colour sensor capable of detecting and quantifying basic colours—Red, Green, and Blue—along with their mixtures. The sensor operates within a supply voltage of -9V to 9V and is able to output corresponding RGB values based on the amount of reflected light from a target surface. This report documents the motivation, design methodology, implementation steps, circuit simulations, calibration techniques, and final testing of the colour sensor. Through careful circuit partitioning and use of voltage scaling, we enable accurate mapping of sensor voltages to RGB values. This low-cost sensor can be integrated with an Arduino and RGB LED strip to visually replicate the sensed colour, forming the basis for potential applications in colour detection systems.**

I. Introduction

Human vision is inherently sensitive to light and colour due to specialized photoreceptors in the eyes—namely rods and cones. Rods are primarily responsible for detecting brightness levels (intensity), while cones detect colour based on the wavelength of incoming light. There are three types of cones in the human eye, each sensitive to a different part of the visible spectrum: red, green, and blue.

In nature, the perception of an object's colour depends on the wavelengths it reflects when illuminated by white light. For example, an apple appears red because it reflects red wavelengths while absorbing the rest. Inspired by this biological principle, we set out to design an electronic system capable of mimicking this behaviour—i.e., detecting how much of the red, green, and blue components are reflected by a surface and converting that into measurable voltage levels.

Our project thus revolves around the concept of using light-dependent sensors to capture reflected light intensities and converting them into RGB values using analog circuitry and calibration. The sensor can detect bright primary colours and estimate mixtures of them, which are further visualized using a programmable RGB LED strip.

II. Working Circuit and Its Components

The overall sensor circuit is divided into three main modules, each responsible for a specific function in the sensing and signal processing pipeline:

1) **Sensor Module** – Detects reflected light intensity for individual colour channels (R, G, B).
2) **Voltage Divider and Isolation Module** – Converts sensor resistance into voltage and prevents interaction between channels.
3) **Voltage Mapping Module** – Amplifies and scales the sensor output voltage into a standard range (0–5V) suitable for digital processing.

A. Sensor Module

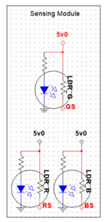

Fig. 1: Sensor Module

The sensor module forms the core of the system. It consists of three discrete colour sensors, each built from a pair of components: a coloured LED and a Light Dependent Resistor (LDR). Each sensor is enclosed within a 2 cm black hollow cube to isolate it from ambient and neighbouring light, ensuring that only the reflected light from the test surface is detected.

Each LED (red, green, or blue) emits light toward the surface, and the corresponding LDR detects the intensity of the reflected light of that wavelength. A higher intensity of reflected light (matching the LED's colour) results in lower resistance of the LDR, while low intensity yields higher resistance. These varying resistances are then processed by the next module to generate meaningful voltage outputs.

B. Voltage Divider and Isolation Module

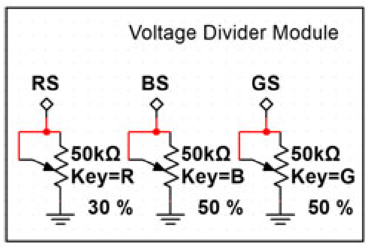

Fig. 2: Voltage Divider Module

The LDR's resistance changes with incident light, but microcontrollers or amplifiers cannot directly interpret resistance changes. Hence, we use voltage divider circuits to convert this resistance into measurable voltage. Each LDR is connected in series with a potentiometer that is manually adjusted to fix the voltage levels at known light intensities (black and white reference).

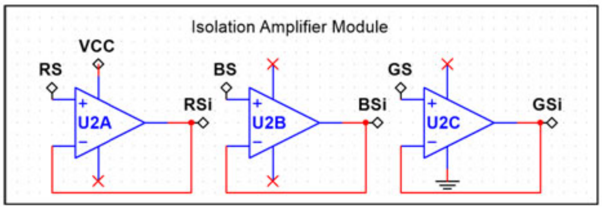

Fig. 3: Isolation Amplifier Module

To ensure signal consistency and prevent electrical loading effects, the output of each voltage divider is buffered using a unity-gain op-amp. This stage ensures that the signal is not distorted or interfered with by other components connected in parallel.

Colour	V_{min} (Black)	V_{max} (White)	Potentiometer
Red	1.5V	2.72V	30%
Green	1.5V	2.1V	5%
Blue	1.5V	2.51V	40%

TABLE I: Measured voltage outputs after initial calibration

C. Voltage Mapping Module

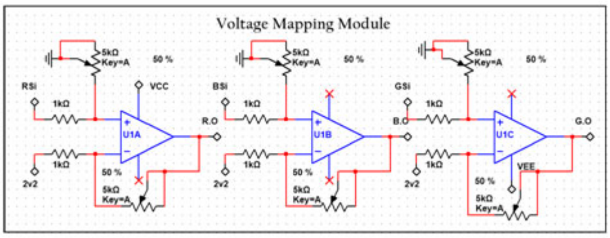

Fig. 4: Voltage Mapping Module

This module takes the buffered analog voltage (typically in the 1.5V–2.7V range) and linearly maps it to a full-scale range of 0V to 5V. The purpose of this mapping is to normalize the sensor outputs and make them compatible with the 0–5V analog input range of a microcontroller such as Arduino.

We use a differential amplifier configuration built using op-amps. The gain of this amplifier is set via a potentiometer (R_G), allowing us to scale different ranges of input to a fixed output range. The reference voltage (V_{min}) is subtracted to shift the base of the signal to 0V.

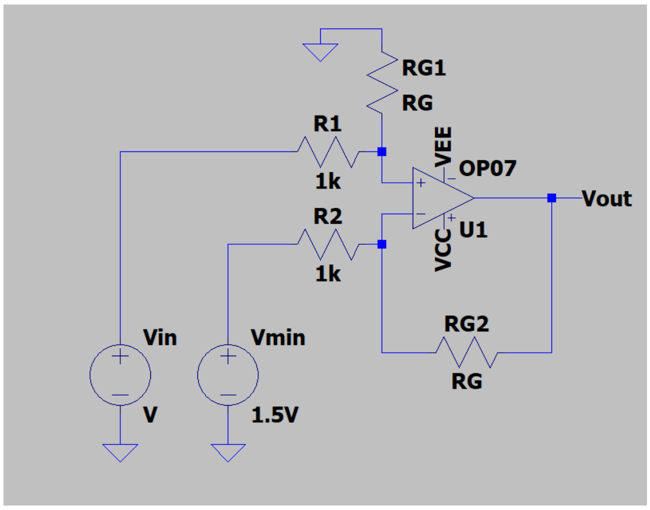

Fig. 5: Circuit for Voltage Mapper

Calculation Derivation: We know from the propertis of opAmp, $V_+ = V_-$. Also, let $R_1 = R_2 = R$. Applying KCL at +ve node,

$$V_+ = \frac{V_1 R_G}{R + R_G}$$

Applying KCL at -ve node,

$$\frac{V_{out} - V_-}{R_G} = \frac{V_- - V_{min}}{R}$$

$$\frac{V_{out}}{R_G} = V_- \left(\frac{1}{R_G} + \frac{1}{R}\right) - \frac{V_{min}}{R}$$

Substitution $V_- = V_+$,

$$\frac{V_{out}}{R_G} = \frac{V_{in} R_G}{R + R_G} \left(\frac{R + R_G}{R_G R}\right) - \frac{V_{min}}{R}$$

$$V_{out} = \left(\frac{R_G}{R}\right)(V_{in} - V_{min})$$

III. Experimentation and Observations

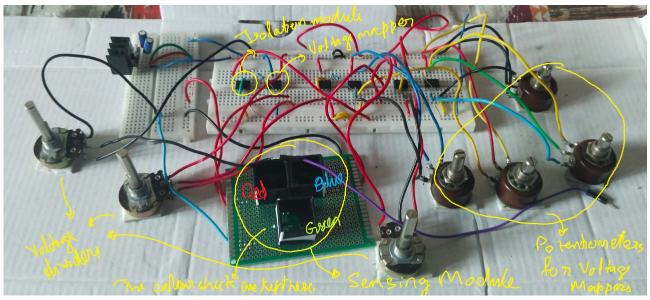

Fig. 6: Final Assembled Circuit

During experimentation, we used a fixed reference voltage of 1.5V (black detection) and set $R = 1k\Omega$ for all differential amplifier circuits. Each colour channel was carefully calibrated by adjusting R_G to produce an output voltage of 5V for white and 0V for black detection.

Colour	Min at VD	Max at VD	Min at VM	Max at VM	Gain at VM
Red	1.5V	2.72V	0V	5.02V	4.114
Green	1.5V	2.1V	0V	2.98V	4.97
Blue	1.5V	2.51V	0V	5.001V	4.95

TABLE II: Calibrated voltage readings and gain values

An Arduino reads these voltages and uses them to drive an RGB LED strip. The LED output closely matches the colour of the sensed object. A demonstration video is available below:

Video Link: Click here to watch

IV. Problems Faced During Building the Circuit

Throughout the design and testing process, several practical challenges emerged:

- **Green LED Intensity Issues:** Unlike red and blue LEDs, the green LED used had significantly lower brightness. As a result, the intensity of reflected light was weaker, leading to a smaller difference between the black and white detection voltages. Consequently, the gain required for green was very high, which made the mapping more sensitive to noise and harder to stabilize at 5V.
- **Change in Reference Voltage:** Initially, we planned to use 2.2V as the minimum reference voltage (V_{min}) for black detection. However, due to insufficient light intensity and reflection in the green channel, we had to reduce the reference voltage to 1.5V. This required recalibration of all sensors and amplifier circuits.
- **Sensitivity to Ambient Light and Material:** Darker coloured objects or matte surfaces reflected less light, which reduced the voltage change and made detection inaccurate. Additionally, the sensor setup was highly sensitive to ambient lighting conditions and even small changes in alignment or object distance affected the output.
- **Calibration Drift Over Time:** The analog components, particularly the LDRs and potentiometers, exhibited drift in behavior over time and temperature. This necessitated recalibration every 30–45 minutes to maintain accuracy. The lack of digital feedback also made precise adjustments difficult.
- **Difficulties in Achieving Pure Colour Output:** Although the system could distinguish colour types (e.g., red from blue), reproducing exact shades on the RGB LED strip proved difficult. Variations in LED output characteristics, minor sensor noise, and imperfect gain tuning contributed to approximate colour matching rather than precise replication.

Conclusion

In this project, we successfully designed and implemented a basic colour sensor that can detect and distinguish between primary colours and their combinations. While there are some limitations in terms of precision and calibration stability, the project demonstrates a working prototype that mimics the human eye's RGB detection system using analog electronics.

With the use of simple components like LEDs, LDRs, op-amps, and potentiometers, we were able to build a modular system that senses and maps colour information, providing real-time output via an RGB LED. This system can be further improved by incorporating digital calibration, higher-quality sensors, and signal conditioning to achieve more accurate and reliable colour recognition.

Overall, the project provided valuable insights into sensor interfacing, analog signal processing, and the challenges of real-world circuit design.